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Phenonium-Ion-Mediated Skeletal Editing of Paracyclophanes

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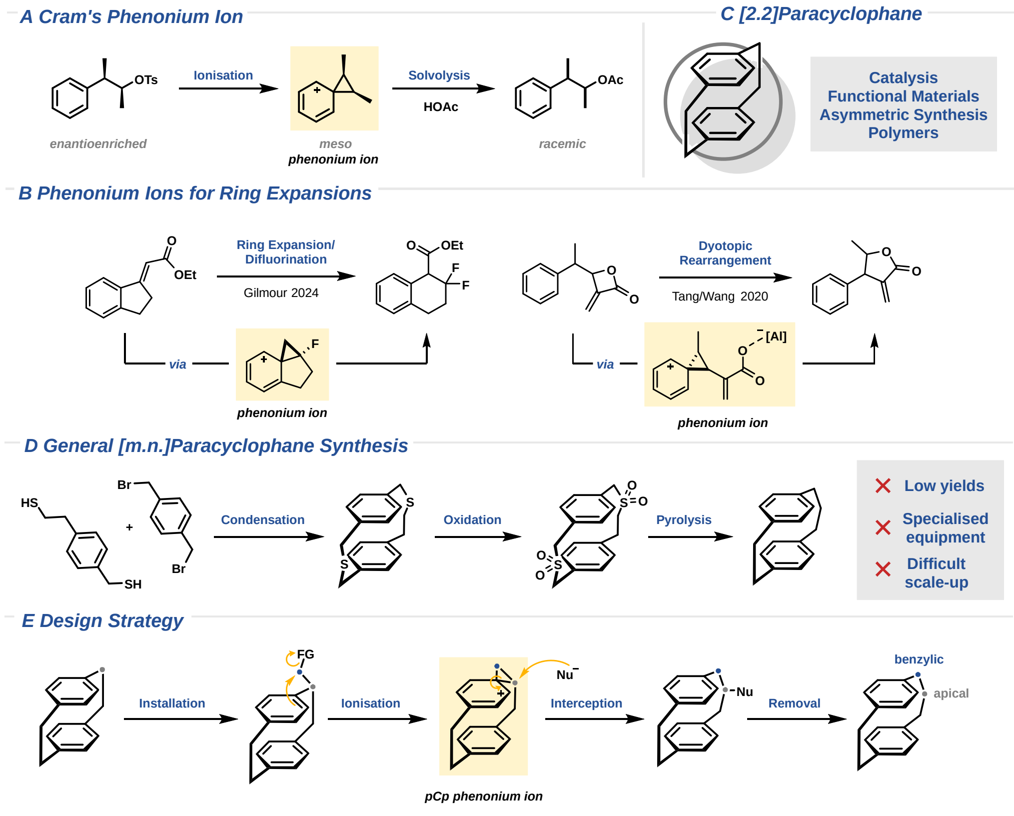
Abstract: Phenonium ions have played a central role in physical-organic chemistry, yet their use as strategic intermediates in synthesis remains underexplored. Here, we report a phenonium-ion-mediated strategy for the skeletal editing of paracyclophanes. Installation of a one-carbon handle at the aliphatic bridge, followed by direct alcohol activation with $BF_3 \cdot OEt_2$ under solvolysis-avoiding conditions, generates a paracyclophane-derived phenonium ion that undergoes regioselective ring opening by external nucleophiles. This strategy provides modular access to functionalised higher paracyclophane homologues and tolerates S-, N-, C-, selenium-, and halide-based nucleophiles under operationally simple, gram-scalable conditions. The sequence can be applied iteratively and extended to double bridge expansion, while reductive desulfurization furnishes the corresponding parent hydrocarbons. Mechanistic studies combining NMR spectroscopy, kinetic analysis, stereochemical probes, and DFT calculations support a self-accelerating mechanism in which product water shifts BF_3 speciation towards $BF_3 \cdot H_2O$, which assists leaving-group departure through hydrogen-bond donation and thereby facilitates phenonium-ion formation. Together, these results establish paracyclophane-derived phenonium ions as controllable intermediates for skeletal editing and provide practical access to higher paracyclophane topologies with application potential in functional materials, as exemplified by a bridge-expanded TADF emitter.

Introduction



Scheme 1: ***(A)*** *Phenonium ion in Cram's seminal solvolysis studies.* ***(B)*** *Recent examples of phenonium-ion-mediated ring expansion.* ***(C)*** *Representative applications of the [2.2]paracyclophane scaffold.* ***(D)*** *Conventional access to higher paracyclophanes via dithiacyclophanes and sulfur extrusion.* ***(E)*** *This work: strategic use of paracyclophane-derived phenonium ions for skeletal editing.*

Phenonium ions occupy a distinctive place in carbocation chemistry. First invoked by Cram in 1949 to rationalise the stereochemical outcome of solvolytic Wagner–Meerwein rearrangements through aryl anchimeric assistance, phenonium ions soon became closely intertwined with the classical/nonclassical carbocation debate (Scheme 1A).[1–3] ^{13}C NMR studies by Olah in 1971 ultimately settled this question, establishing phenonium ions as classical, spirocyclic carbocations.[4–7] However, despite their prominent role in physical-organic chemistry, phenonium ions have only gradually been harnessed as strategic intermediates in organic synthesis.[8]

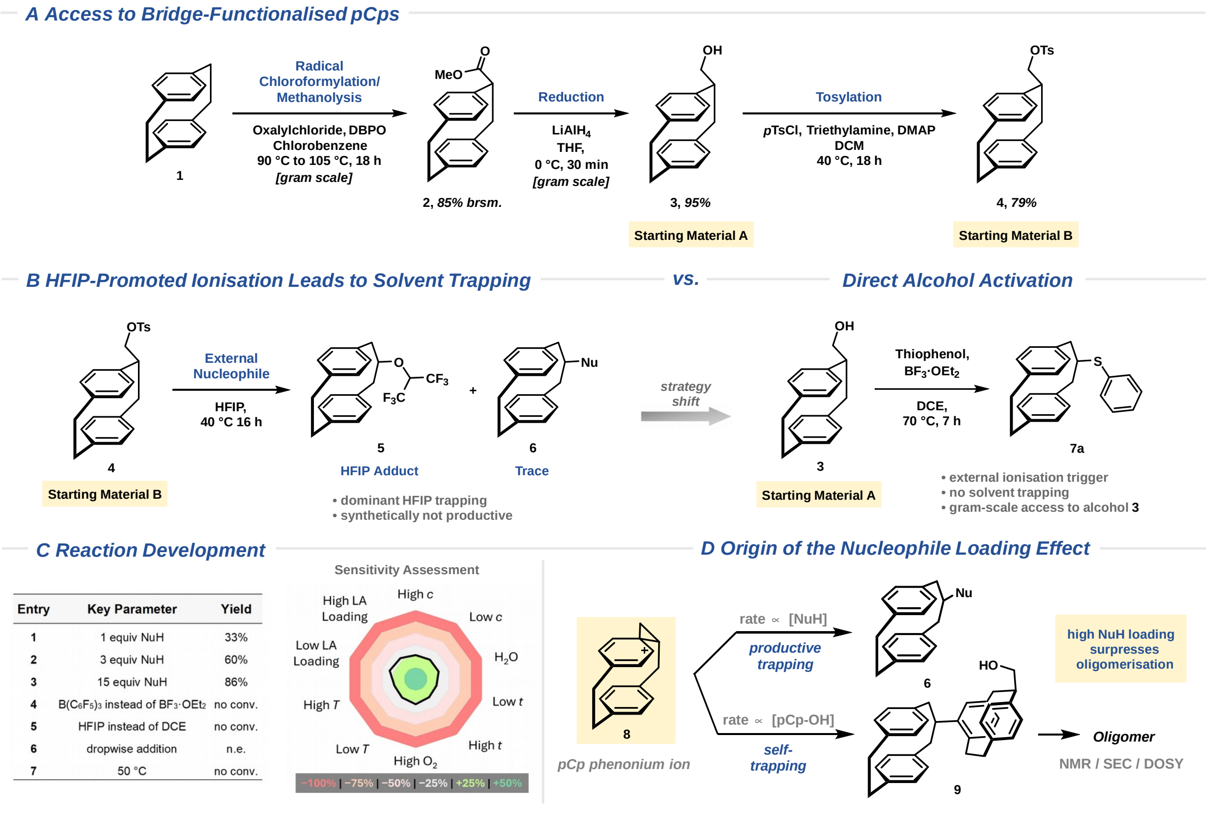
From a synthetic perspective, the central challenge is taming the high intrinsic reactivity of these cationic species. Early studies largely relied on activated substrates, typically featuring good leaving groups and electron-rich aryl systems, whereas recent advances, including hexafluoroisopropanol (HFIP)-enabled ionisation of 2-arylethanols, reagent-controlled regioselective opening of unsymmetrical phenonium ions, and hypernucleofuge-based strategies using hypervalent iodine reagents, have broadened the substrate scope of phenonium-ion chemistry and demonstrated that increasing levels of synthetic control over both cation formation and interception can be achieved through judicious reaction design.[9–15] Importantly, studies from the groups of Gilmour and Tang/Wang have shown that phenonium ions can be used as synthetic tools for the ring expansion of cyclic scaffolds (Scheme 1B).[16,17] Beyond these isolated examples, however, the use of phenonium ions as a design principle for skeletal editing remains underappreciated. Herein, this concept is translated to the skeletal editing of paracyclophanes, using pCp-derived phenonium ions to modularly access higher cyclophane homologues.

Paracyclophanes (pCps) are strained hydrocarbon scaffolds in which two cofacially stacked aromatic decks are held in close proximity by aliphatic bridges. Among them, [2.2]paracyclophanes are the most widely studied representatives, with two ethylene bridges locking the aromatic decks into a rigid, distorted arrangement (Scheme 1C). This constrained geometry promotes through-space π-interactions between the decks and, upon suitable substitution, can give rise to planar chirality. Together, these features have made [2.2]pCps valuable motifs in catalysis, asymmetric synthesis, polymer chemistry, and functional materials.[18–23]

Although arene-deck functionalisation has enabled broad diversification of the [2.2]pCp scaffold, methods for modifying the ethylene bridges remain far less advanced.[24,25] Still more limited are approaches that edit the bridge architecture itself, despite the fact that bridge length directly controls strain, interdeck distance, molecular geometry, and transannular electronic communication. The ability to modulate these parameters would therefore be highly attractive for diverse applications, yet the synthetic access needed to explore higher and unsymmetrically bridged pCp homologues in this context is severely restricted.[26,27] Today, access to the latter mainly relies on high-dilution condensation to form dithia- or sulfone-bridged cyclophanes, followed by extrusion of sulfur (photochemically) or SO_2 (*via* flash-vacuum pyrolysis) (Scheme 1D). These routes are limited by the low-yielding condensation step and by the practical constraints of specialised equipment, which precludes routine use and complicates scale-up.[26,28,29] Consequently, operationally simple, scalable, and modular strategies for homologating readily available [2.2]paracyclophanes into higher cyclophane frameworks are still needed, making pCps an attractive but challenging target class for skeletal editing.[30] Prior studies of pCp-derived phenonium ions, however, were conducted in the context of solvolysis experiments, where solvent capture of the cationic intermediate offered little opportunity for synthetic diversification.[31–34] We sought to develop a strategy that would suppress solvent trapping and redirect the pCp-derived phenonium ion towards selective interception by external nucleophiles.

The presented strategy relies on the selective installation of a one-carbon handle at the ethylene bridge of [2.2]paracyclophane, followed by phenonium ion formation and nucleophilic interception to achieve the desired ring expansion (Scheme 1E). Overall, the introduced carbon atom is rearranged into the cyclophane bridge and occupies the benzylic position, whereas the nucleophile that traps the cation becomes attached at the apical position of the expanded bridge. Subsequent removal of the functional handle can furnish the corresponding cyclophane homologue. Realising this design therefore required identifying conditions that would convert a historically difficult-to-control cationic intermediate into a synthetic tool for skeletal editing.

Results and Discussion



Entry	Key Parameter	Yield
1	1 equiv NuH	33%
2	3 equiv NuH	60%
3	15 equiv NuH	86%
4	$B(C_6F_5)_3$ instead of $BF_3{\cdot}OEt_2$	no conv.
5	HFIP instead of DCE	no conv.
6	dropwise addition	n.e.
7	50 °C	no conv.

Scheme 2: ***(A)*** *Synthesis of bridge-functionalised [2.2]paracyclophane precursors.* ***(B)*** *Strategic shift from tosylate ionisation in HFIP to direct alcohol activation to avoid solvolysis.* ***(C)*** *Reaction development and sensitivity assessment; isolated yields are given. Abbreviations: LA = Lewis acid, c = concentration, t = reaction time, T = temperature, H_2O = added water, and O_2 = oxygen.* ***(D)*** *Self-trapping of the pCp phenonium ion accounts for the nucleophile-loading effect.*

The required bridge functionality was introduced by radical chloroformylation of [2.2]paracyclophane (**1**)[35,36] followed by reduction of the resulting pCp ester **2** to the alcohol **3** (Starting Material A), which was subsequently transformed to the corresponding pCp tosylate **4** (Starting Material B) (Scheme 2A).

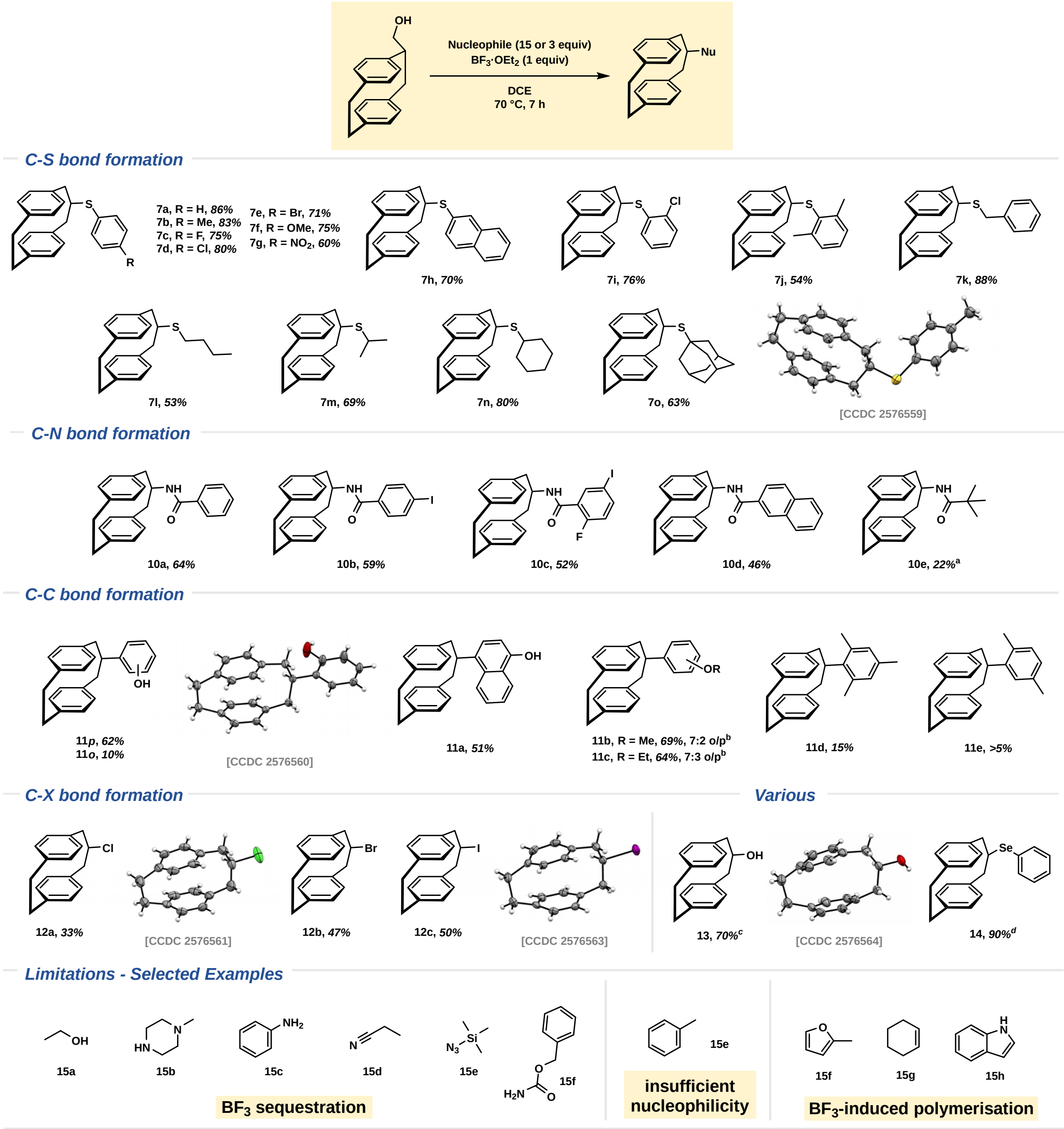
Conceptually guided by recent reports on HFIP-based phenonium-ion chemistry,[15] we first examined whether the corresponding pCp tosylate **4** (Starting Material B) could be ionised in HFIP (Scheme 2B). Indeed, ionisation was observed under these conditions; however, the resulting pCp-derived phenonium ion was rapidly intercepted by the weakly nucleophilic solvent, delivering the HFIP adduct **5** as the major product (see Section S3.6). This outcome highlighted the key challenge for a synthetically useful transformation: the conditions had to enable the formation of the high-energy phenonium intermediate while avoiding solvolysis, and to channel the intermediate towards trapping by an external nucleophile. We therefore turned to direct activation of pCp alcohol **3** (Starting Material A) under nonpolar conditions. This precursor is readily available in two steps on multigram scale and allows the ionisation to be triggered by an external Lewis or Brønsted acid. Using DCE as solvent and thiophenol as a model nucleophile, we evaluated a broad set of acids. Most candidates either failed to activate the starting material or led to unproductive pathways (see Section S3.7). $BF_3{\cdot}OEt_2$ was uniquely effective, giving full conversion and the rearranged [3.2]paracyclophane thioether **7a** in an initial 33% yield. Other boron Lewis acids, such as $B(C_6F_5)_3$, did not promote the transformation under otherwise identical conditions (Scheme 2C, entry 4). Although phenonium-ion generation from alcohols has literature precedent under HFIP/TfOH or superacidic conditions, such reactivity under nonpolar conditions has, to the best of our knowledge, not been reported.[15,37]

Reaction optimisation identified nucleophile loading as the key parameter: with the model nucleophile, the yield increased from 33% with 1 equivalent (Scheme 2C, entry 1) to 60% with 3 equivalents (entry 2) and 86%

with 15 equivalents (entry 3). No conversion was observed at 50 °C (entry 7). HFIP alone was ineffective, and combining HFIP with $BF_3 \cdot OEt_2$ also led to no conversion, presumably because the Lewis acid is sequestered by the alcohol solvent (entry 5). A Glorius-type sensitivity assessment[38] showed that the transformation tolerates air and small deviations in concentration or temperature, while prolonged reaction times of up to 40 h did not erode the yield, indicating that the thioether product is stable under the reaction conditions. In contrast, both lower and higher Lewis acid loadings had a negative effect on the outcome. Unexpectedly for a transformation initiated by $BF_3 \cdot OEt_2$, the reaction also proved somewhat tolerant to water, an observation that would later become central to the mechanistic analysis (see Section S3.8).

We next examined why the reaction depends so strongly on nucleophile loading (Scheme 2D). When only one equivalent of thiophenol was used, an additional side-product fraction was isolated alongside the desired [3.2]pCp thioether **7a**. This material was not detected under the optimised conditions, indicating that it arises from a competing pathway that becomes relevant at low nucleophile concentration. NMR and DOSY NMR spectroscopy, together with SEC analysis, indicated that the side product is not a discrete compound, but an oligomeric mixture composed of pCp-derived units. A related material also formed in the absence of an external nucleophile. On this basis, we attribute the competing pathway to repeated Friedel–Crafts-type self-trapping of the phenonium ion **8** by the starting pCp alcohol **3**, leading to a pCp-based oligomeric mixture. Thus, although lower nucleophile loadings still afforded synthetically useful yields, the use of a large nucleophile excess can be mechanistically rationalised by the need to outcompete Friedel–Crafts-type self-trapping of the cationic intermediate **8**. Following this rationale, we examined whether slow addition of pCp alcohol **3**, intended to keep its concentration low and thereby suppress oligomerisation, would improve the yield (Scheme 2C, entry 6). However, this setup had no beneficial effect (see Section S3.9).

Based on these studies, we defined two sets of conditions. For thiol nucleophiles, 15 equivalents were used to maximise yield, since excess reagent could be readily removed by a simple basic aqueous wash. In contrast, non-thiol nucleophiles were employed in three equivalents, which provided a compromise between yield and ease of purification. In both cases, the reaction was performed with 1.0 equivalent of $BF_3 \cdot OEt_2$ in DCE at 70 °C and did not require dry solvent or an inert atmosphere, making the setup operationally straightforward.



Scheme 3: Nucleophile scope investigation. Reactions were performed on a 0.5 mmol scale. Isolated yields are given. [a] 3.00 equivalents of BF_3·OEt_2 were used. [b] Combined yield of both regioisomers is given. o/p ratio was determined by ^{13}C NMR. [c] 3.00 equivalents of allyltrimethylsilane and 36 h reaction time were used. [d] 5.00 equivalents of benzeneselenol were used.

With an optimised protocol in hand, we investigated the range of nucleophiles capable of intercepting the pCp-derived phenonium ion under these conditions (Scheme 3). Aryl thiols bearing electron-donating and electron-neutral substituents all afforded the corresponding [3.2]paracyclophane thioethers in high yields, ranging from 71% to 86%, with electron-poor 4-nitrobenzenethiol furnishing **7g** in 60% yield. Substitution at the aryl thiol was also well tolerated, including ortho-substitution (**7i**, 76%) and extended aromatic systems (**7h**, 70%). The reaction further accommodated benzylic (**7k**, 88%), linear aliphatic (**7l**, 53%), branched aliphatic (**7m**, 69%), and cyclic (**7n**, 80%; **7o**, 63%) thiols, giving the corresponding thioethers in moderate to high yields.

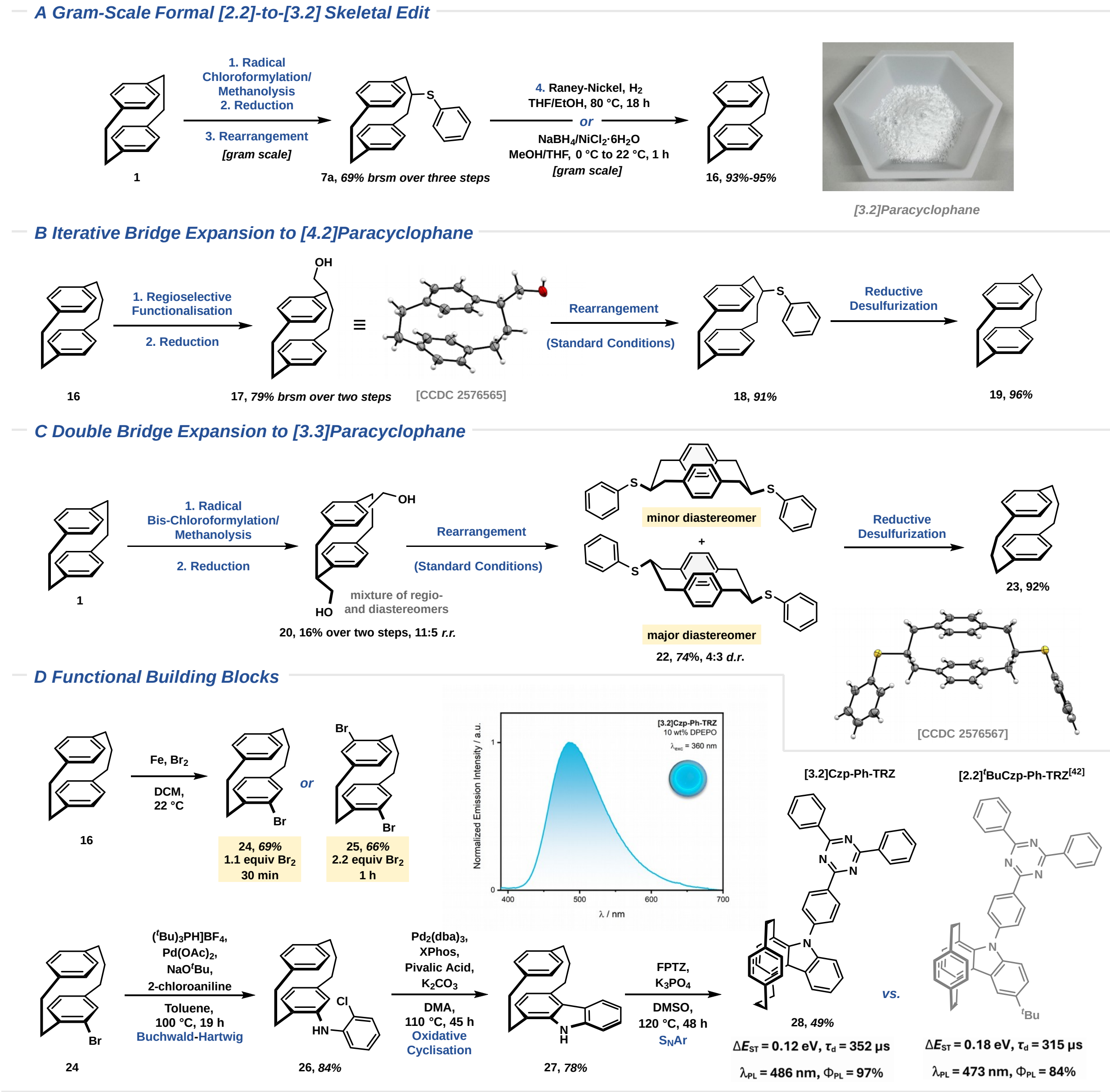
The scope of non-thiol nucleophiles was then evaluated using the three-equivalent protocol as described above. Under these more demanding conditions, nitrogen-based trapping proceeded through a Ritter-type manifold with aryl nitriles to furnish the corresponding amide products (**10a–10d**, 46% to 64%). Alkyl nitriles were generally not tolerated, presumably because they sequestered the Lewis acid more efficiently than aryl nitriles. A notable exception was pivalonitrile, which delivered the corresponding amide **10e** only when the BF_3·OEt_2 loading was increased to three equivalents in a yield of 22%.

Carbon-based trapping was possible but showed a clear dependence on the nucleophilicity of the arene partner. Phenolic nucleophiles reacted via C–C rather than C–O bond formation, likely because coordination of the phenolic oxygen to $BF_3{\cdot}OEt_2$ deactivates the oxygen centre (**11a**, 51%; **11*o*/*p***, 72% combined yield). Electron-rich anisole (**11b**, 69%) and phenetole (**11c**, 64%) intercepted the cationic intermediate in moderate yields, whereas the less nucleophilic mesitylene gave the corresponding product **11d** in a substantially lower 15% yield. The behaviour of 1,4-xylene was particularly informative: it provided only trace amounts of the trapping product **11e** alongside the oligomeric side product, whereas less electron-rich toluene led exclusively to oligomerisation. These observations suggest that xylene lies near the electronic threshold required for external arene trapping to compete with self-trapping by the pCp alcohol **3**.

Halide incorporation was achieved using TMS halide sources, providing access to the corresponding chloro-, bromo-, and iodo-substituted [3.2]paracyclophanes (**12a–12c**, 33–50%). Allyltrimethylsilane (allyl-TMS) showed distinct reactivity. Instead of furnishing the allylated [3.2]pCp product, the reaction initially provided a mixture of the starting [2.2]pCp alcohol **3** and the rearranged [3.2]pCp alcohol **13**. Extending the reaction time to 36 h led to full conversion and afforded the [3.2]pCp alcohol **13** in 70% yield. In contrast, the use of only one equivalent of allyl-TMS resulted in oligomerisation. Although the exact role of allyl-TMS in this process remains unclear, these observations suggest that it acts as a reaction additive, enabling interception of the phenonium ion by the oxygen atom of the leaving group generated upon ionisation (see Section S3.10). Notably, selenoether **14** was obtained in 90% yield using reduced loading of 5 equivalents of benzeneselenol, highlighting the high trapping efficiency of this softer nucleophile.

Importantly, downstream functionalisation of the apical bridge position in [3.2]paracyclophane derivatives is severely limited, since classical S_N2 displacement is precluded by the inaccessible backside trajectory. This limitation underscores the value of introducing different nucleophiles during the phenonium-ion-mediated rearrangement itself, since it provides direct access to products that cannot be obtained by simple post-functionalisation of, for example, [3.2]paracyclophane iodide **12c**.

The limitations uncovered during our studies fall into three categories: (1) strongly Lewis-basic nucleophiles (e.g. amines, alcohols, anilines), which sequester $BF_3{\cdot}OEt_2$ and thereby prevent ionisation altogether; (2) weakly nucleophilic arenes, which do not interfere with ionisation but fail to outcompete self-trapping once the pCp phenonium ion has formed (e.g. toluene); and (3) nucleophiles prone to Lewis-acid-induced polymerisation or decomposition (e.g. thiophenes, furans, alkenes). Thus, productive trapping requires a balance between nucleophilicity, Lewis basicity, and stability under the reaction conditions.



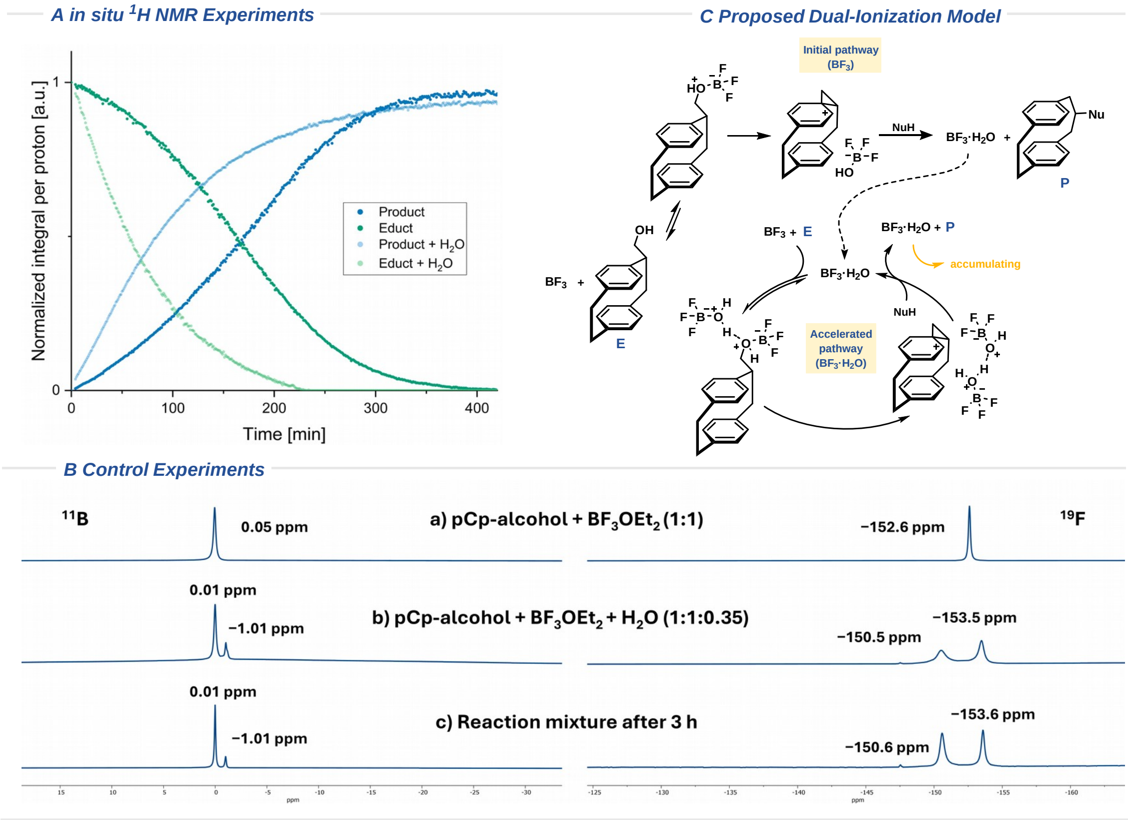
*Scheme 4: **(A)** Gram-scale formal skeletal editing of [2.2]paracyclophane to [3.2]paracyclophane. **(B)** Iterative bridge expansion. **(C)** Double bridge expansion. **(D)** Synthetic elaboration of [3.2]paracyclophane (**16**) towards TADF emitter **28** (left), photoluminescence spectrum of the 10 wt% emitter-doped DPEPO film (λ_{exc} = 360 nm, centre), and comparison of the photophysical properties with the corresponding ^{t}Bu[2.2]carbazolophane[42] emitter (right).*

The [3.2]paracyclophane phenyl thioether **7a** was readily converted into the parent [3.2]paracyclophane hydrocarbon **16** by reductive desulfurization using either Raney-Nickel/H_2 or $NaBH_4/NiCl_2{\cdot}6H_2O$,[39] giving the hydrocarbon in high yield (93% or 95%, respectively) (Scheme 4A). This transformation completes the formal [2.2]-to-[3.2] skeletal edit and provides a straightforward four-step access to the mixed-bridge [3.2] hydrocarbon framework from [2.2]paracyclophane (**1**). Furthermore, the whole sequence could be readily translated to the gram scale.

We next asked whether the phenonium-ion-mediated bridge expansion could be applied iteratively, that is, whether an already expanded [3.2]paracyclophane (**16**) could undergo a second bridge expansion (Scheme 4B). Unlike the symmetric ethylene bridge of [2.2]paracyclophane (**1**), the three-carbon bridge of the [3.2]pCp hydrocarbon **16** offers two distinct positions for functionalisation. Applying the chloroformylation-methanolysis/reduction sequence to this species regioselectively installed the hydroxymethyl group at the three-carbon bridge, as confirmed by single-crystal X-ray diffraction analysis of alcohol **17**. This selectivity is consistent with the more pronounced benzylic character of the functionalised position in the three-carbon bridge compared with the corresponding positions in the ethylene bridge. Subjecting alcohol **17** to the standard rearrangement conditions afforded the [4.2]paracyclophane thioether **18** in 91% isolated yield, and subsequent reductive desulfurization furnished the parent [4.2]paracyclophane (**19**) in 96% yield. Thus, the bridge-expansion sequence can be iteratively extended to higher-order paracyclophane topologies.

A bis-functionalised [2.2]paracyclophane substrate was prepared by treating the [2.2]pCp starting material **1** with increased amounts of the chloroformylation reagents and prolonged reaction time (Scheme 4C). This transformation delivered a mixture of regio- and diastereomers, as confirmed by X-ray structures of the corresponding alcohols **20** (see Section S3.11). Subjecting this mixture to the standard conditions led to efficient double bridge expansion, affording the corresponding [3.3]paracyclophane bis-thioether **22** as a mixture of diastereomers in 74% combined yield. The minor diastereomer could be selectively crystallised, and its structure was confirmed by X-ray diffraction. Subsequent reductive desulfurization of the diastereomeric bis-thioether mixture **22** then furnished the corresponding [3.3]paracyclophane hydrocarbon **23** in 92% yield.

Because bridge length in paracyclophanes directly affects molecular geometry, strain, interdeck distance, and transannular electronic communication, higher paracyclophane homologues represent attractive building blocks for functional molecular architectures.[26,27] To demonstrate that the bridge-expanded scaffolds obtained by this methodology can be further elaborated, we investigated the functionalisation of parent [3.2]paracyclophane (**16**) (Scheme 4D). Electrophilic bromination provided mono- or dibrominated [3.2]paracyclophanes (**24** or **25**), while in the monobromination, substitution occurred preferentially on the arene deck adjacent to the shorter ethylene bridge.[27] Building on our recent work on [2.2]carbazolophane-based thermally activated delayed fluorescence (TADF) emitters,[40–42] we used monobrominated [3.2]paracyclophane **24** as an entry point to a bridge-expanded carbazolophane donor. Buchwald–Hartwig coupling with 2-chloroaniline followed by oxidative cyclisation furnished donor **27**, which was subsequently coupled with 2-(4-fluorophenyl)-4,6-diphenyl-1,3,5-triazine (FPTZ) to give the donor–acceptor emitter **28**. Photophysical characterisation in a 10 wt% doped film in DPEPO revealed sky-blue emission at λ_{PL} of 486 nm with a photoluminescence quantum yield, Φ_{PL}, of 97% under nitrogen (85% in air), representing a significant improvement over the corresponding [2.2]carbazolophane emitters.[40–42] Relative to the previously reported [2.2]^{t}BuCzp-Ph-TRZ benchmark,[42] the bridge-expanded emitter exhibits slightly higher S_1 and T_1 levels, increasing from 2.87 to 2.93 eV for S_1 and from 2.69 to 2.81 eV for T_1, respectively, resulting in a reduced ΔE_{ST} of 0.12 versus 0.18 eV, although the delayed emission lifetime slightly increased from 315 to 352 µs, likely due to reduced non-radiative decay. Overall, these results demonstrate that bridge topology serves as an effective design parameter for cyclophane-based functional molecular architectures (see Section S3.12).



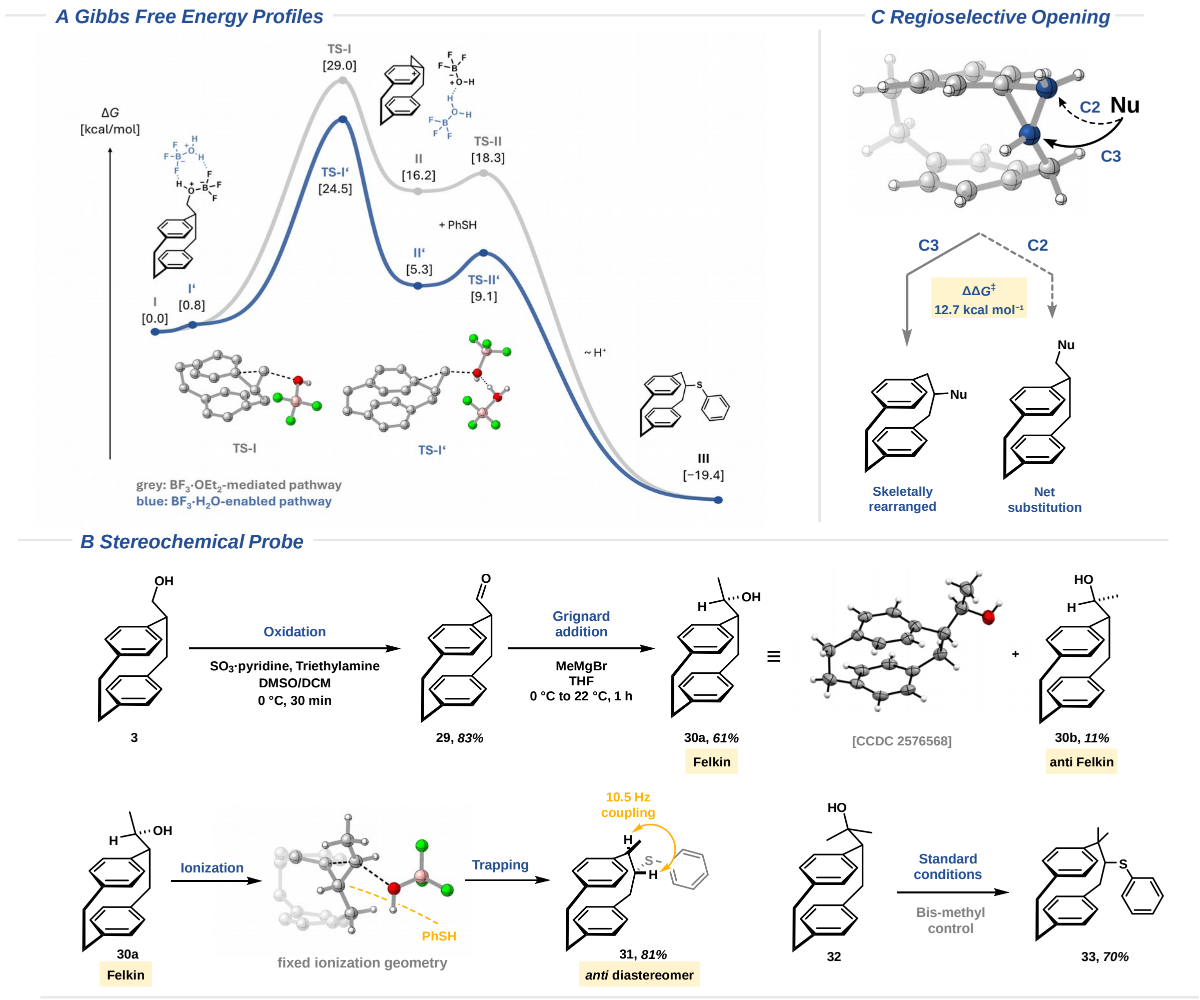
*Scheme 5: **(A)** In-situ ^{1}H NMR spectroscopy experiment reveals sigmoidal kinetic profile. **(B)** Control experiments. **(C)** Proposed dual-ionisation model.*

The unexpected tolerance of the reaction towards water suggested that the reaction warranted closer mechanistic examination. Accordingly, an in-situ ^{1}H NMR spectroscopy experiment was conducted to monitor the reaction kinetics (Scheme 5A). Interestingly, the starting-material and product profiles displayed an induction period and sigmoidal kinetics, suggesting a self-accelerating process (see Section S3.13).[43–45]

To identify the origin of this behaviour, we noted that water is generated during the reaction from the alcohol substrate and the protic nucleophile and hypothesised that water-induced changes in BF_3 speciation could account for the observed self-acceleration. In particular, we considered the in-situ formation of $BF_3 \cdot H_2O$, which is described in the literature as a Brønsted acid and for which rate-accelerating effects in BF_3-mediated reactions have been documented.[46–52] To test whether $BF_3 \cdot H_2O$ is indeed formed under the reaction conditions, we conducted ^{11}B and ^{19}F NMR spectroscopy control experiments (Scheme 5B). Mixing the pCp alcohol **3** with $BF_3 \cdot OEt_2$ gave a single ^{11}B resonance at $\delta_B = 0.05$ ppm. Addition of 0.35 equivalents of water generated a second signal at $\delta_B = -1.01$ ppm, which was also observed in a reaction mixture sampled after three hours. The ^{19}F NMR experiments gave analogous results. After water addition, a second ^{19}F resonance at $\delta_F = -150.5$ ppm appeared and the initial resonance shifted from $\delta_F = -152.6$ ppm to $\delta_F = -153.5$ ppm, with both signals reproduced in the sampled reaction mixture. Furthermore, in-situ ^{11}B and ^{19}F NMR spectroscopy experiments confirmed the gradual formation of this species during the reaction (see Section S3.13). Notably, Zhang reported ^{11}B NMR spectral shifts for $BF_3 \cdot H_2O$ in very good agreement with our data, while excluding other boron species such as HBF_4 and H_3BO_3.[48] Accordingly, based on our control experiments and the literature precedent, we assign the newly formed species to $BF_3 \cdot H_2O$ and conclude that it forms progressively over the course of the reaction.

To determine whether the in-situ formation of $BF_3 \cdot H_2O$ affects the kinetic profile, an additional ^{1}H NMR spectroscopy experiment was performed under otherwise identical conditions in the presence of 0.35 equivalents of added water (Scheme 5A). Compared with the original experiment, the induction period was effectively eliminated and both substrate consumption and product formation were accelerated. These findings strongly support the involvement of $BF_3 \cdot H_2O$ in the observed self-acceleration.

The ionisation is expected to be the rate-determining step in this transformation. Together with the kinetic and control experiments described above, this consideration led us to propose a dual ionisation model (Scheme 5C). Within this model, at the start of the reaction, $BF_3 \cdot OEt_2$ promotes slow ionisation of the pCp alcohol **E**, generating small amounts of thioether product **P** and water. The water formed in this process shifts BF_3 speciation towards $BF_3 \cdot H_2O$, which opens an alternative, lower-barrier pathway for phenonium-ion formation. Continued water release during product formation increases the concentration of $BF_3 \cdot H_2O$, which, together with the lower ionisation barrier, accounts for the observed induction period and self-acceleration. This dual role of BF_3, acting both as an initial Lewis acid activator and as the precursor to the more active $BF_3 \cdot H_2O$ species, may also rationalise the unique performance of $BF_3 \cdot OEt_2$ observed during the optimisation studies.



*Scheme 6: **(A)** Calculated Gibbs free-energy profiles for the $BF_3 \cdot OEt_2$-mediated pathway (grey) and the $BF_3 \cdot H_2O$-enabled pathway (blue). Calculations were performed at the SMD(1,2-dichloroethane)/ ωB97X-D4/def2-TZVP//CPCM(1,2-dichloroethane)/ωB97X-D4/def2-TZVP level using Garza–Gellrich solution-entropy corrections.[53,54] Relative Gibbs free energies are given in kcal mol^{-1} and referenced to I + $BF_3 \cdot H_2O$ + PhSH. **(B)** Stereochemical probe supports stereospecific phenonium-ion formation and trapping. **(C)** Regioselectivity of phenonium-ion ring opening.*

While this mechanistic model accounts for the observed kinetic profile, it does not establish how exactly $BF_3 \cdot H_2O$ promotes ionisation at the molecular level. We therefore used DFT calculations to explore plausible molecular mechanisms. The initial pathway involves Lewis-acid-mediated ionisation, whereas we first interpreted the $BF_3 \cdot H_2O$-enabled pathway as Brønsted-acid-mediated, in line with the literature precedent.[48–52] However, the results discussed below point to a more intricate chemical scenario.

As expected, the calculated Gibbs free-energy profile for the $BF_3 \cdot OEt_2$-mediated pathway shows that ionisation is energetically highly demanding, with **TS-I** located at $\Delta G^{\ddagger}$ = 29.0 kcal mol^{-1} relative to the BF_3-activated pCp alcohol **I** (Scheme 6A). The transition geometry shows simultaneous elongation of the C–O bond and formation of the new C–C bond in a nearly collinear arrangement, consistent with the classical S_N2-

type description of phenonium-ion formation, in which leaving-group departure and C–C bond formation occur in a concerted fashion.[55] Subsequent interception of the phenonium ion by thiophenol is nearly barrierless ($\Delta G^{\ddagger}$ = 2.1 kcal mol^{-1}), and formation of the [3.2]pCp thioether **III** is strongly exergonic relative to **I** (ΔG = −19.4 kcal mol^{-1}).

Central to the proposed self-acceleration model is whether $BF_3{\cdot}H_2O$ enables a lower-energy pathway for phenonium-ion formation. Notably, all attempts to locate a stationary point corresponding to a discrete protonated pCp alcohol were unsuccessful. Although a concerted protonation–ionisation pathway was identified, its effective barrier of 29.1 kcal mol^{-1} is essentially identical to that of the $BF_3{\cdot}OEt_2$-mediated pathway (see Section S5.3). Taken together, these findings argue against a classical Brønsted-acid mechanism. Instead, the calculations revealed a cooperative $BF_3/BF_3{\cdot}H_2O$ pathway. In the reactant complex **I′**, $BF_3{\cdot}H_2O$ interacts through a cyclic pair of O–H⋯F hydrogen bonds with the BF_3-complex **I**. Upon formation of **TS-I′**, the alcohol oxygen accepts a hydrogen bond from $BF_3{\cdot}H_2O$ which facilitates C–O bond cleavage by stabilising the departing trifluoro(hydroxy)borate moiety. This cooperative activation lowers the effective ionisation barrier to 24.5 kcal mol^{-1}, 4.5 kcal mol^{-1} below that of the $BF_3{\cdot}OEt_2$-mediated pathway. The resulting phenonium intermediate **II′** is also substantially lower in free energy than **II** on the common reference scale ($\Delta\Delta G$ = −10.9 kcal mol^{-1}). Thus, in our system, $BF_3{\cdot}H_2O$ is best understood not as a conventional Brønsted acid, but as a hydrogen-bond donor that facilitates leaving-group departure within a Lewis-acid-activated complex.

We next used the stereochemical behaviour of substituted alcohol substrates as an experimental probe (Scheme 6B). Oxidation of the alcohol **3** to the corresponding pCp aldehyde **29**, followed by diastereoselective addition of methylmagnesium bromide, furnished the Felkin alcohol **30a** as the major product together with the anti-Felkin alcohol **30b**. The relative configuration of the major diastereomer **30a** was assigned by single-crystal X-ray diffraction. Subjecting the Felkin alcohol **30a** to the standard rearrangement conditions afforded the corresponding [3.2]paracyclophane thioether **31** as a single *anti*-diastereomer. The *anti* relationship between the apical proton and the benzylic proton was established by coupling-constant analysis (see Section S3.14). In contrast, the anti-Felkin alcohol **30b** decomposed under the same conditions, and the origin of this divergent behaviour could not be conclusively established. However, it cannot be explained by steric hindrance, since the corresponding bis-methyl alcohol **32** was successfully converted in 70% yield. The exclusive formation of *anti*-configured product **31** is consistent with the calculated transition-state geometry and supports a stereospecific rearrangement.

Finally, beyond stereospecificity, the use of the pCp-derived phenonium ion as a strategic intermediate for skeletal editing also relies on a second defining feature of its reactivity: highly regioselective ring opening (Scheme 6C). Controlling regioselectivity is central for phenonium-ion-mediated transformations, and recent work by Race has made important progress in addressing this challenge through reagent-controlled ring opening.[13] In the present system, nucleophilic attack could in principle occur at either C2 or C3. While C2 attack would lead to net substitution, C3 attack gives the bridge-expanded framework. Calculations show that C3 attack is favoured over C2 attack by a substantial $\Delta\Delta G^{\ddagger}$ of 12.8 kcal mol^{-1}, accounting for the highly regioselective, substrate-controlled outcome observed experimentally.

Conclusion

In summary, this work establishes a modular phenonium-ion-mediated strategy for the skeletal editing of paracyclophanes. Synthetic control over the pCp-derived phenonium ion is achieved through direct alcohol activation under nonpolar conditions that avoid solvolysis, using the dual-role Lewis acid $BF_3{\cdot}OEt_2$ that initiates formation of the cationic intermediate and serves as the precursor for the in-situ formation of $BF_3{\cdot}H_2O$, which further promotes ionisation. Under these conditions, external nucleophiles can outcompete self-trapping and oligomerisation when present in sufficient concentration and compatible with the Lewis acidic medium. The intrinsic regioselectivity of the pCp-derived phenonium ion channels the reactivity towards bridge expansion rather than net substitution. Under operationally simple conditions, this strategy tolerates a diverse range of nucleophiles and provides modular access to [3.2]paracyclophanes. Reductive desulfurization delivers the parent hydrocarbons, and the sequence can be translated to gram scale, applied iteratively to higher homologues, and extended to double bridge expansion. The resulting higher paracyclophane topologies serve as functional building blocks, as exemplified by the incorporation into a TADF emitter architecture. Together, these results define pCp-derived phenonium ions as controllable intermediates for skeletal editing and provide a practical entry into higher paracyclophane homologues for synthesis and materials-oriented applications.

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